3 Shear Force and Bending Moment

3.1 Introduction

A beam is a structural member subjected to a system of external forces at right angles to its axis. The process of deformation of a bar because of the forces transverse to its axis is called bending. The algebraic sum of the vertical forces at any section of a beam to the right or left of the section is known as shear force. It is briefly written as S.F. The algebraic sum of the moments of all the forces acting to the right or left of the section is known as bending moment. It is written as B.M. idered.

3.2 Shear Force And Bending Moment Diagrams

A shear force diagram is one which shows the variation of the shear force along the length of the beam. A bending moment diagram is one which shows the variation of the bending moment along the length of the beam.

3.3 Types Of Loads

A beam is normally horizontal and the loads acting on the beams are generally vertical. The following are the important types of load acting on a beam:

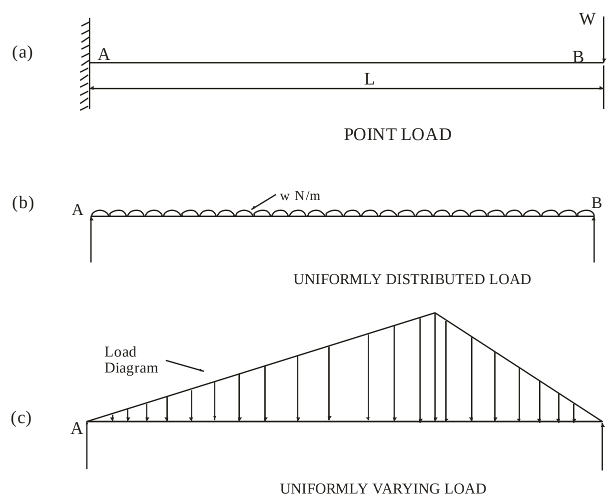

Fig. 3.1

3.3.1. Concentrated Or Point Load. A concentrated load is one which is, considered to act at a point. Although in practice it must really be distributed over a small area because such knife edge contacts are neither possible nor desirable. In Fig. 3.1(a) W shows the point load acting on a beam.

3.3.2. Uniformly Distributed Load. A uniformly distributed load is one which is spread over a beam in such a manner that intensity of loading *w* is uniform along the length *(i.e.* each unit length is loaded to the same rate) as shown in Fig. 3.1(b). The intensity of loading is expressed as *w* N/m run.

3.3.3. Uniformly Varying Load. A uniformly varying load is one which is spread over a beam in such a manner that intensity loading varies from point to point along the beam as shown in Fig. 3.1(c) in which load is zero at one end and increases uniformly to one point and then decreases to zero to the other end.

3.4 Types Of Beams

The following are the important types of beams:

3.4.1. Cantilever Beam. A beam which is fixed at one end and free at the other end is known as cantilever beam. Such beam is shown in Fig. 3.2(a).

3.4.2. Simply Supported Beam. A beam supported or resting freely on the supports at its both ends is known as simply supported beam. Such beam is shown in Fig. 3.2(b).

3.4.3 Overhanging Beam. If the end portion of a beam is ex-tended beyond the support, such beam is known as overhanging beam. Overhanging beam is shown in Fig. 3.2(c).

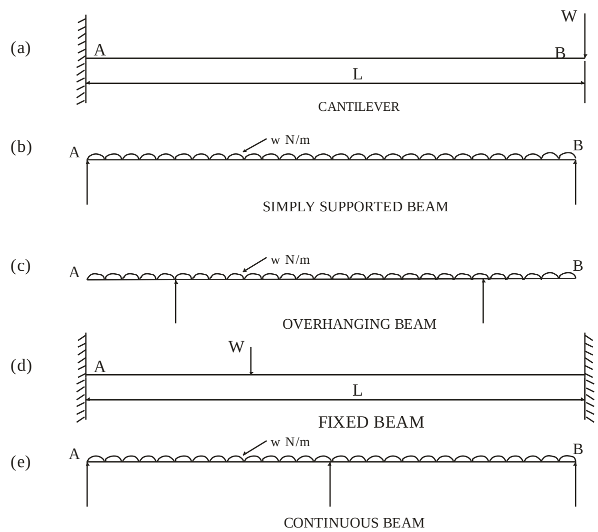

Fig. 3.2

3.4.4. Fixed Beams. A beam, whose both ends are fixed or built in walls, is known as fixed beam. Such beam is shown in Fig. 3.2(d). A fixed beam is also known as a *built-in*, or encastred beam.

3.4.5. Continuous Beam. A beam which is provided more than two supports or more than one span as shown in Fig. 3.2(e) is known as continuous beam.

3.5 Shear Force and Bending Moment

Fig. 3.3(a) shows a simply supported of span L. A single point load Pacts at a distance of L_1 from left support. The two reactions at left and right supports may be calculated from equations of static equilibrium. These equations state that the algebraic sum of all forces acting along two mutually perpendicular directions is zero. In case of beams one direction is taken along the beam axis and second direction is transverse to the beam axis. The equations of equilibrium further state that algebraic sum of moments of all forces acting on a body taken about any point in the plane of forces will vanish.

Taking moment about right support:

$$R_1L = W(L-L_1) \text{ or } R_1 = \frac{W(L-L_1)}{L}$$

Summing up the forces transverse to the beam axis

$$R_2 - W + R_1 = 0 \text{ or } R_1 = \frac{WL_1}{L}$$

Where R_1 & R_2 are the support reactions.

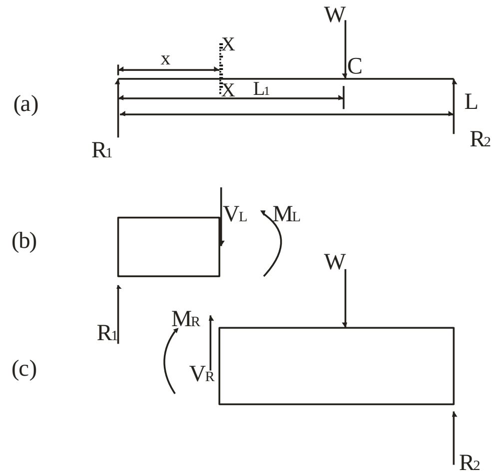

Fig. 3.3

Fig. 3.3(b) shows the free body diagram of left hand portion of the beam cut by a transverse plane XX at a distance of x from left support. It can be seen that to balance R_1on the right side of the cut portion a downward internal force V_L equal and opposite to R_1 ($V_L = - R_1$) will act on the section at a distance x from the left support. The force V_L is called shear force. R_1 and V_L will form a clockwise couple R_1x and hence an internally developed moment i.e. moment of resistance ($M_L = -R_1x$) will have to balance the moment R_1x and will act anticlockwise. (We are taking clockwise moment positive.)

Thus it now becomes clear that in a loaded beam at any section the static equilibrium is achieved by two internal reactions viz a shear force V_L and a moment of resistance M_L. By drawing a free body diagram of right hand portion of the beam as in Fig. 3.3(c) it can be shown that to balance $(-W+R_2)$ on the left side of the cut portion a upward internal force V_R equal and opposite to $(-W+R_2)$ will act on the section.

$$V_R = -(-W+R_2) = +R_1$$

The force V_R is called shear force. $-(-W+R_2)$ and V_R will form a anticlockwise couple:

$$M = W(L_1-x) - \frac{WL_1}{L}(L-x) = -\frac{WL_2}{L}x = -R_1x$$

Hence an internally developed moment i.e. moment of resistance ($M_R = R_1x$) will have to balance the above moment R_1x and will act clockwise. (We are taking clockwise moment positive). We can observe that V_L and V_R are same in magnitude but opposite in signs and acting on opposite sides of the sectioning plane. Similarly M_L and M_R are same in magnitude but opposite in directions.

‘The **shear force** at a section of a loaded beam is the algebraic sum of all the forces acting either left or right side of the section. Similarly **bending moment** at a section is the algebraic sum of moments of all the forces either to the right or to the left of the section taken about the section.’

3.6 Sign Conventions:

(a) Shear Force: shear force having a downward direction to the right hand side of a section will be taken as positive. Similarly shear force having an upward direction to the right hand side of a section will be taken as negative. (Fig. 3.4(a))

(b) Bending Moment: A bending moment causing convexity upwards will be taken as negative and will be called a hogging. Similarly a bending moment causing concavity upwards will be taken as positive and will be called a sagging. (Fig. 3.4(b))

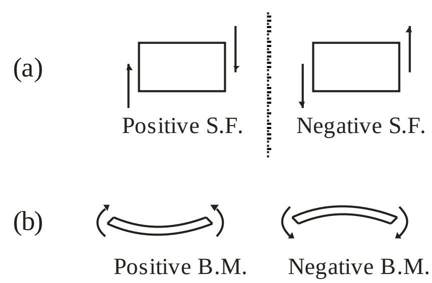

Fig. 3.4

3.7 Important Points For Drawing Shear Force And Bending Moment Diagrams

The shear force diagram is one which shows the variation of the shear force along the length of the beam. A bending moment diagram is one which shows the variation of the bending moment along the length of beam. In these diagrams, the shear force or bending moment are represented by ordinates whereas the length of the beam represents abscissa.

The following are the important points for drawing shear force and bending moment diagrams:

1. Consider the left or the right portion of the section.

2. Add the forces (including reaction) normal to the beam on one of the portion. If right portion of the section is chosen, a force on the right portion acting downwards is positive while a force on the right portion acting upwards is negative.

   If the left portion of the section is chosen, a force on the left portion acting upwards is positive while a force on the left portion acting downwards is negative.

3. The positive values of shear force and bending moments are plotted above the base line, and negative values below the base line.

4. The shear force diagram will increase or decrease suddenly i.e. by a vertical straight line at a section where there is a vertical point load.

5. The shear force between any two vertical loads will be constant and hence the shear force diagram between two vertical loads will be horizontal.

6. The bending moment at the two supports of a simply supported beam and at the free end of a cantilever will be zero.

<u>3.8 Shear Force And Bending Moment Diagrams For A Cantilever</u>

3.8.1 A Cantilever With A Point Load At The Free End

Fig. 3.5 (a) shows a cantilever AB of length L fixed at A and free at B and carrying a point load W at the free end B. Let us take a section X at a distance x from the free end.

Let F_x = Shear force at X, and

M_x = Bending moment at X.

S.F.D: Considering the right portion of the section the shear force at this section is equal to the resultant force acting on the right portion at the given section. The resultant force acting on the right portion at the section X is W which is acting in the downward direction. Since a force on the right portion acting downwards is considered positive, hence shear force at *X* is positive.

$\therefore F_x = +W$

The shear force will be constant at all sections of the cantilever between *A* and *B* as there is no other load between *A* and *B.* The shear force diagram is shown in Fig. 3.5 (b).

B.M.D.

The bending moment at the section *X* is given by

$M_x = -W \times x$ ...(i)

(Bending moment will be negative as B.M. is of hogging nature, bending the beam

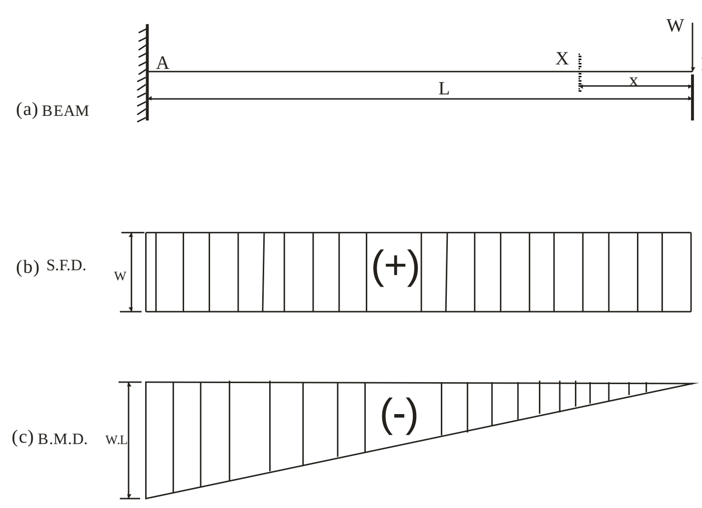

Fig. 3.5

convex upwards.)

From equation *(i)*, it is clear that B.M. at any section is proportional to the distance of the section from the free end.

At *x*=0 *i.e.* at *B*, B.M.= 0

At *x*=*L* *i.e.* at *A*, *B.M.*=*W*×*L*

Hence B.M. follows the straight line law. The B.M. diagram is shown in Fig. 3.5 (c).

<u>3.8.2 A Cantilever With Several Point Loads</u>

S.F.D.: Between B and D, F_x = + 4 (constant)

Between D and C, F_x = + 4 + 4 = 8 (constant)

Between C and A, F_x = + 4 + 4 + 2 = 10 (constant)

Therefore the S.F.D. consists of rectangles of different sizes as in Fig. 3.6 (b).

B.M.D.: Between B and D, M_x = - 4x

At B, x = 0 $\therefore$ M_B = 0.

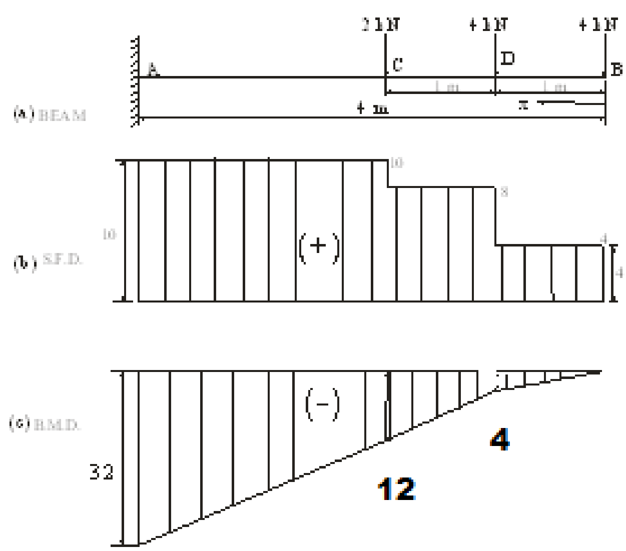

Fig. 3.6

At D, x = 1 m $\therefore$ M_D = - 4x1 = -4 kN-m.

Between D and C, M_x = - 4x – 4 (x – 1)

At D, x = 1 m $\therefore$ M_D = -4 kN-m.

At C, x = 2 m $\therefore$ M_C = -12 kN-m.

Between C and A, $M_x = -4x - 4(x-1) - 2(x-2)$

At C, $x = 2$ m $\therefore M_C = -12$ kN-m.

At A, $x = 4$ m $\therefore M_A = -32$ kN-m.

Therefore the B.M.D. consists of lines of various slopes as in Fig. 3.6 (c).

3.8.3 A Cantilever With A Uniformly Distributed Load

Fig. 3.7 shows a cantilever of length *L* fixed at *A* and carrying a uniformly distributed load of w per unit length over the entire length of the cantilever.

S.F.D: The shear force at any section *X* at a distance x from the free end B will be equal to the resultant force acting on the right portion of the section. This resultant force is acting downwards. Resultant force on the right portion acting downwards is considered positive. Hence shear force at *X* is positive.

The resultant force on the right portion = w × Length of right portion = $+w.x$.

At B, $x = 0 \therefore F_B = 0$.

At A, $x = L \therefore F_A = +wL$.

Therefore the S.F.D. consists of triangle as shown in Fig. 3.7(b).

B.M.D.:

To find the effect of the uniformly distributed load over a section it is converted into point load acting at the C.G. of the uniformly distributed load.

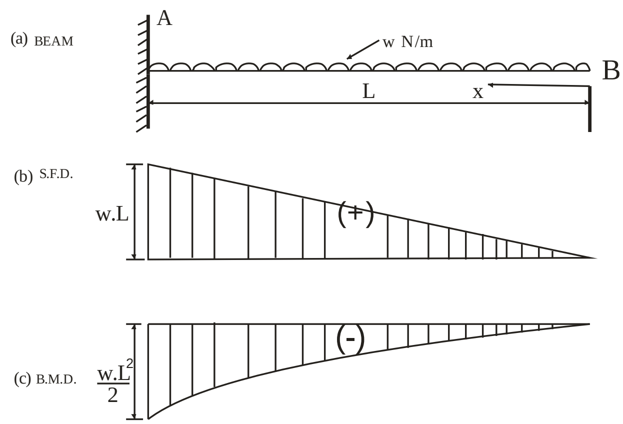

Fig. 3.7

The bending moment at the section *X* is given by

M_x=-(*Total* load on right portion) × Distance of section from the C.G. udl

$$= -(w \cdot x) \cdot \frac{x}{2}$$

$$= -w \cdot x \cdot \frac{x}{2} = -w \cdot \frac{x^2}{2} \qquad \text{.... (i)}$$

(The bending moment will be negative as for the right portion of the section, the moment of the load at x is clockwise. Also the bending of cantilever will take place in such a manner that convexity will be at the top of the cantilever).

From equation *(i)*, it is clear that B.M. at any section follows a parabolic law.

At B, $x = 0$ hence $M_B = 0$

At A, $x = L$ hence $M_A = - w \cdot \frac{L^2}{2}$

The bending moment diagram is drawn as shown in Fig. 3.7(c).

3.8.4 A Cantilever With A U.D.L. On Part Span From The Free End

S.F.D:

For section BC, $F_x = +w.x$

At B, $x = 0 \therefore F_B = 0$.

At A, $x = a \therefore F_C = +w.a$.

For section CA, $F_x = +w.a$ (Constant)

At A, $x = L \therefore F_A = F_C = +w.a$.

The S.F.D. is triangular from B to C and rectangular from C to A as shown in Fig. 3.8 (b).

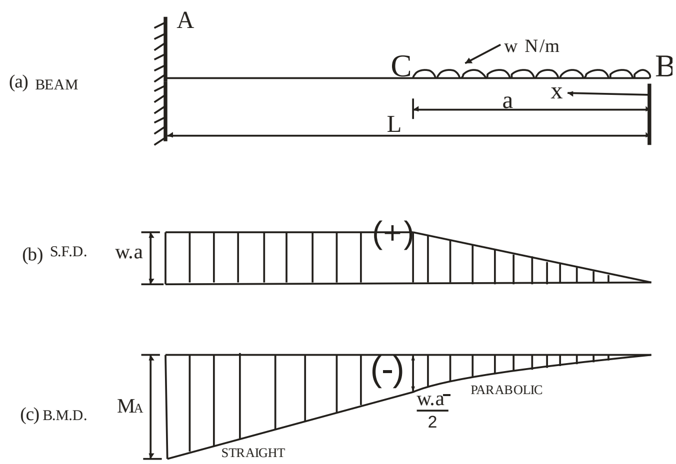

Fig. 3.8

B.M.D.: For section BC, $M_x = - w. \frac{x^2}{2}$ (Parabolic)

At B, $x = 0 \therefore M_B = 0$.

At C, $x = a \therefore M_C = - w. \frac{a^2}{2}$.

For section CA, $M_x = - wa. \left(x - \frac{a}{2}\right)$ (linear)

At C, $x = a \therefore M_C = -w. \frac{a^2}{2}$.

At A, x = L $\therefore$ $M_A = -\,w.a\left(L-\frac{a}{2}\right)$.

The B.M.D. is parabolic from B to C and linear from C to A as shown in Fig. 3.8 (c).

3.8.5 A Cantilever With Combination of Loads (U.D.L. and Point loads)

S.F.D:

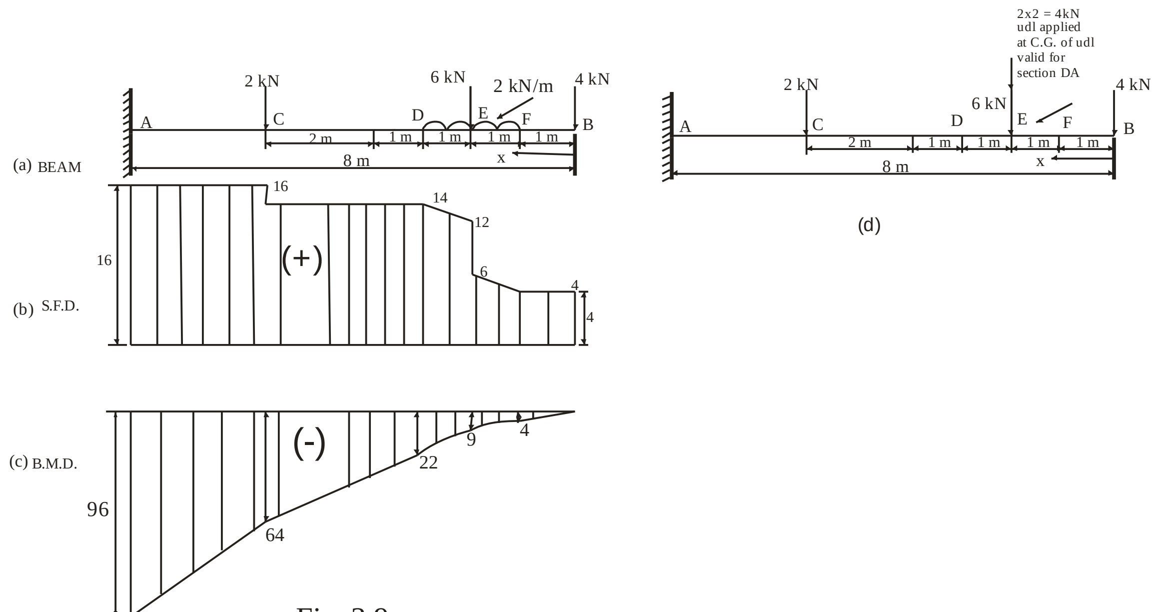

Fig. 3.9

For section BF, F_x = +4 kN (Constant)

At B, x = 0 $\therefore$ F_B = +4 kN

At F, x = 1 $\therefore$ F_F = +4 kN

For section FE, F_x = +4+2.(x-1)

At F, x = 1 $\therefore$ F_F = +4 kN

At E, x = 2 $\therefore$ F_E = +6 kN

For section ED, F_x = +4+2.(x-1)+6

At E, x = 2 $\therefore$ F_E = +1 kN. (Sudden change in S.F. due to point load)

At D, x = 3 $\therefore$ F_D = +14 kN

For section DC, F_x = +14 kN (Constant)

At D, x = 3 $\therefore$ F_D = +14 kN

At C, x = 6 $\therefore$ F_C = +14 kN

For section CA, F_x = +14+2 = 16 kN (Constant)

At C, x = 6 $\therefore$ F_C = +16 kN (Sudden change in S.F. due to point load)

At A, x = 8 $\therefore$ F_A = +16 kN

The S.F.D. is shown in Fig. 3.9 (b).

B.M.D.: For section BF, M_x = 4.x

At B, x = 0 $\therefore$ M_B = 0.

At F, x = 1 $\therefore$ M_F = - 4 kN-m

For section FE, M_x = - 4.x - $\frac{2}{2}(x-1)^2$ (Parabolic)

At F, x = 1 $\therefore$ M_F = - 4 kN-m

At E, x = 2 $\therefore$ M_E = - 9 kN-m

For section ED, M_x = - 4.x - $\frac{2}{2}(x-1)^2$ - 6 (x-2) (Parabolic)

At E, x = 2 $\therefore$ M_E = - 9 kN-m

At D, x = 3 $\therefore$ M_D = - 22 kN-m

For section DC, This section is out of the udl. Hence the udl is to be converted into equivalent point load. The total udl (2x2) is assumed to be applied at C.G. of the udl i.e. 2 m away from the free end as shown in Fig. 3.9 (d). To draw BMD for section DA Fig. 3.9 (d) is to be considered.

M_x = - 4.x – 4.(x-2)- 6 (x-2) (Linear)

At D, x = 3 $\therefore$ M_D = - 22 kN-m

At C, x = 6 $\therefore$ M_C = - 64 kN-m

For section CA

M_x = - 4.x – 4.(x-2)- 6.(x-2)- 2.(x-6) (Parabolic)

At C, x = 6 $\therefore$ M_C = - 64 kN-m

At A, x = 8 $\therefore$ M_A = - 96 kN-m

The B.M.D. is shown in Fig. 3.8 (c).